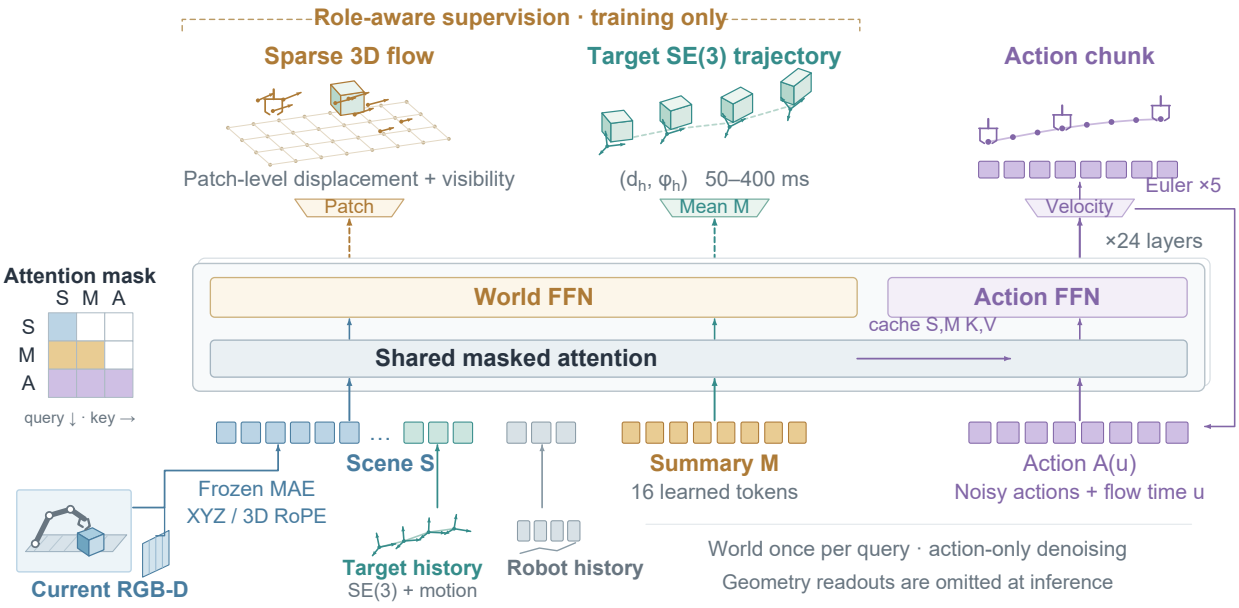

Role-aware supervision · training only
Sparse 3D flow
Patch-level displacement + visibility
Patch
Target SE(3) trajectory
$(d_h, \varphi_h)$ 50–400 ms
Mean M
Action chunk
Euler ×5
Velocity
×24 layers
Attention mask
S M A
S
M
A
query ↓ · key →
World FFN
Action FFN
cache S,M K,V
Shared masked attention
Scene S
Summary M
16 learned tokens
Action A(u)
Noisy actions + flow time u
Frozen MAE
XYZ / 3D RoPE
Current RGB-D
Target history
SE(3) + motion
Robot history
World once per query · action-only denoising
Geometry readouts are omitted at inference